Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | McLean, VA area (local - no relocation)

PROFESSIONAL SUMMARY

AI agent builder who **ships production agentic systems end-to-end** - the core of the Agentforce AI Builder role. Built AgentProbe from scratch: a **3-agent LangGraph platform** (Orchestrator, RAG Tracer, Auto Fix) with tool calls, multi-step reasoning, and inter-agent state handoffs - the same architecture Agentforce agents use for enterprise deployments. **Multi-provider LLM routing** (Anthropic, OpenAI, Groq, Ollama), **5-metric eval engine** with CI/CD gates, and a **15-endpoint FastAPI backend** - full-stack, production-grade, open-source. **8+ years production Python** + TypeScript/React/GraphQL at Entrata (100K+ MAU, 1.6M records, 120+ Agile sprints). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). Local to McLean VA - available immediately. *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

AI Agent Architecture (LangGraph, Tool Calls) | LLM Integration & Prompt Engineering | Python (8+ years production)

React / TypeScript / GraphQL | Multi-Provider LLM Routing | FastAPI / REST Platform Layer

RAG Pipelines & Vector Search (ChromaDB) | LLM Evaluation & Observability | Enterprise-Scale Delivery (100K+ MAU)

Docker / AWS / CI/CD | End-to-End Project Ownership | Cross-Functional Technical Communication

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Took full ownership of **20+ full-stack modules** (Python, TypeScript/JavaScript, React, PostgreSQL, GraphQL) across **120+ Agile sprints** - delivered end-to-end for **100K+ monthly active users**, partnering with engineers, product managers, and business teams to translate requirements into production systems - the same customer-to-code pattern Agentforce AI Builders execute.
- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI on regulated enterprise data, the domain Agentforce serves.
- Led Docker-based containerization for 30%+ of microservices, cutting execution time by **50%**; improved production query performance by **30%**; maintained **99.5% uptime** across enterprise SaaS.
- Migrated client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents**; built 5+ platform tools adopted across squads, reducing resolution time by **40%**.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia (local)

- Translated complex technical concepts into working solutions for 30+ students - same communication pattern required when partnering with Salesforce customers to transform business challenges into agentic deployments.

PROJECTS

AgentProbe: AI Agent Platform - The Open-Source Agentforce Pattern OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · TypeScript · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built the same system Agentforce AI Builders deploy for enterprise customers - production-grade AI agents from architecture to deployment. **Agent architecture**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) with multi-step reasoning, tool calls, failure handling, and inter-agent state handoffs - the exact pattern Agentforce agents use. **LLM stack**: multi-provider routing across Anthropic, OpenAI, Groq, Ollama with dynamic selection by task and capability; ChromaDB vector store + embeddings + retrieval quality scoring. **Evaluation**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD gates - agents are reliable and auditable in production. **Platform**: 15-endpoint FastAPI backend + React observability dashboard. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: Healthcare Enterprise AI - LLMs + FHIR APIs (HIPAA-Compliant)

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · GraphQL

End-to-end enterprise AI integration for a regulated domain - NLP routing natural language queries to FHIR APIs via LLMs, BERT, and embeddings. **HIPAA-compliant RBAC. ~60% retrieval time reduction.** Demonstrates the full-stack enterprise AI deployment pattern Agentforce AI Builders execute for major companies. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, Tool Calls, LLM APIs (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Vector Search, LLM-as-Judge Evaluation, CI/CD Eval Gates, Prompt Engineering, Context Engineering

Frontend & Backend: React, FastAPI, Flask, GraphQL, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum, Claude Code, GitHub Copilot, Cursor